TU Dortmund University
Department of Statistics

# Application Report

M.Sc. Data Science

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## Student Performance Analysis:

## Gender and Parental Education Effects on Mathematics and Language Test Scores

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# 1 Introduction

In every modern society, education is both the great equaliser and a mirror of structural inequality. Governments invest heavily in schooling with the expectation that talent and effort — not gender or family background — should determine academic outcomes. Yet decades of evidence challenge this ideal: systematic differences in Mathematics and Language achievement persist along lines of gender and parental education across many countries Else-Quest2010,Davis-Kean2005. Quantifying these gaps, verifying their statistical significance, and identifying which sub-populations drive them is therefore scientifically important and practically urgent for educators and policymakers alike.

The first predictor of interest is *gender*. Cross-national meta-analyses document a recurrent pattern: male students tend to outperform female students in Mathematics, while female students outperform male students in verbal and language tasks, with effect sizes that are consistently small-to-moderate Else-Quest2010. These differences are not merely statistical curiosities; they shape course selection, career trajectories, and long-term wage outcomes. The second predictor is *parental level of education*, a widely used proxy for the cultural capital and material resources available in the home. Higher parental education is consistently associated with richer learning environments, greater academic expectations, and better access to educational support Davis-Kean2005, translating into measurable test-score advantages for children.

The dataset examined here contains $n = 486$ student records with four variables: `gender` (nominal, binary), `parental_education` (ordinal, four ordered levels: high school, associate degree, bachelor's degree, master's degree), `math_score` and `language_score` (both ratio-scaled integers, [0, 100], no missing values). Full-sample averages are $\bar{x}_{\text{math}} = 68.9$ ($s = 13.0$) and $\bar{x}_{\text{lang}} = 69.6$ ($s = 13.5$), indicating broadly comparable difficulty across the two subjects.

Two research questions guide the analysis. **RQ1**: Is there a statistically significant difference between male and female students in their Mathematics and Language scores? **RQ2**: Is there a statistically significant difference in scores across the four parental education levels, and if so, which specific group pairs differ?

The main findings are as follows. Male students score significantly higher in Mathematics ($\bar{x}_{\text{male}} = 71.9$ vs. 65.5, $p < 0.001$, effect size $|r| = 0.25$), while female students score significantly higher in Language ($\bar{x}_{\text{female}} = 72.3$ vs. 67.8, $p = 0.001$, $|r| = 0.19$). For parental education, significant differences are confirmed across all groups, with students from master's-degree households outperforming high-school students by 12.6 points in Mathematics and 10.3 points in Language.

The remainder of the report is structured as follows. Section 2 describes the dataset and research questions in detail. Section 3 defines all statistical methods with mathematical rigour. Section 4 presents descriptive and inferential analyses. Section 5 summarises, interprets, and discusses the findings.

# 2 Problem Description

## 2.1 Research Questions

**RQ1.** Is there a statistically significant difference between male and female students in their Mathematics and Language scores?

**RQ2.** Is there a statistically significant difference in scores across parental education levels? If so, which group pairs differ?

## 2.2 Dataset Description

The dataset contains $n = 486$ observations with no missing values. `gender` is nominal (male: $n = 263$, 54.1%; female: $n = 223$, 45.9%). `parental_education` is ordinal with four ordered levels: high school ($n = 132$, 27.2%), associate degree ($n = 127$, 26.1%), bachelor's degree ($n = 120$, 24.7%), master's degree ($n = 107$, 22.0%). Equal spacing between levels cannot be assumed; the variable is therefore treated as a categorical grouping factor. Table 1 summarises all variables.

Table 1 : *Dataset variable overview ($n = 486$).*

| Variable | Scale | Values | $n$ / Summary |
|---|---|---|---|
| `gender` | Nominal | male / female | 263 / 223 |
| `parental_education` | Ordinal | hs < assoc. < bach. < mast. | 132/127/120/107 |
| `math_score` | Ratio | $[0, 100]$, integer | $\bar{x} = 68.9,\ s = 13.0$ |
| `language_score` | Ratio | $[0, 100]$, integer | $\bar{x} = 69.6,\ s = 13.5$ |

# 3 Statistical Methods

## 3.1 Descriptive Statistics

Let $x_1, \ldots, x_n$ denote $n$ observations. The **arithmetic mean** and **sample variance** Wackerly2008 are:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 \tag{1}$$

Standard deviation $s = \sqrt{s^2}$; interquartile range $\mathrm{IQR} = Q_3 - Q_1$. The **median** $\tilde{x}$ (distinct from $\bar{x}$) Wackerly2008:

$$\tilde{x} = \begin{cases} x_{\left(\frac{n+1}{2}\right)} & n \text{ odd}, \\ \frac{1}{2}\left[x_{\left(\frac{n}{2}\right)} + x_{\left(\frac{n}{2}+1\right)}\right] & n \text{ even}. \end{cases} \tag{2}$$

## 3.2 Graphical Methods

A **box plot** Tukey1977 shows the five-number summary per group; whiskers extend to $Q_1 - 1.5\,\mathrm{IQR}$ and $Q_3 + 1.5\,\mathrm{IQR}$. A **heatmap** encodes matrix values as colour intensity

Waskom2021. The **Pearson correlation coefficient** Wackerly2008:

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\,\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}, \quad r \in [-1, 1]. \tag{3}$$

### 3.3 Hypothesis Tests, Normality Check, and Effect Sizes

Normality is assessed with the **Shapiro-Wilk test** Shapiro1965 ($H_0$: normal; reject when $p < 0.05$):

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}, \quad W \in (0, 1] \tag{4}$$

where $x_{(i)}$ is the $i$-th order statistic and $a_i$ are tabulated constants Shapiro1965. Violations of normality in several sub-groups (Table 5) rule out the parametric $t$-test (RQ1) and one-way ANOVA (RQ2), so non-parametric alternatives are used throughout.

For **RQ1**, the **Mann-Whitney U test** Mann1947 ($H_0$: distributions identical; $\alpha = 0.05$):

$$U_1 = n_1 n_2 + \tfrac{n_1(n_1+1)}{2} - R_1, \qquad U = \min(U_1, U_2) \tag{5}$$

Effect size: **rank-biserial correlation** $r_{\text{rb}} = 1 - \frac{2U}{n_1 n_2}$ Kerby2014; $|r| \approx 0.1$ (small), 0.3 (medium), 0.5 (large).

For **RQ2** ($k = 4$), the **Kruskal-Wallis H test** Kruskal1952:

$$H = \frac{12}{N(N+1)} \sum_{j=1}^{k} \frac{R_j^2}{n_j} - 3(N+1), \quad H \overset{H_0}{\sim} \chi^2_{k-1} \tag{6}$$

Post-hoc: all $\binom{4}{2} = 6$ pairwise Mann-Whitney U tests with **Bonferroni correction** Bland1995 and effect size $\eta_H^2$ Tomczak2014,Cohen1988:

$$\alpha_{\text{adj}} = \frac{0.05}{6} \approx 0.0083, \qquad \eta_H^2 = \frac{H - k + 1}{N - k} \tag{7}$$

$\eta_H^2$: $\approx 0.01$ (small), $\approx 0.06$ (medium), $\geq 0.14$ (large) Cohen1988.

# 4 Evaluation

## 4.1 Descriptive Analysis

### 4.1.1 Scores by Gender

Table 2 shows male students score 6.4 pts higher in Mathematics and female students 4.5 pts higher in Language; spread is similar ($s \approx 14$). Figure 1 confirms the pattern with clearly labelled means, medians, and quartiles.

Table 2 : *Descriptive statistics by gender.*

| Score | Gender | $n$ | $\bar{x}$ | $s$ | $\tilde{x}$ |
|---|---|---|---|---|---|
| Mathematics | Male | 263 | 71.9 | 13.9 | 72.0 |
| | Female | 223 | 65.5 | 14.2 | 65.0 |
| Language | Male | 263 | 67.8 | 13.8 | 67.0 |
| | Female | 223 | 72.3 | 13.1 | 72.0 |

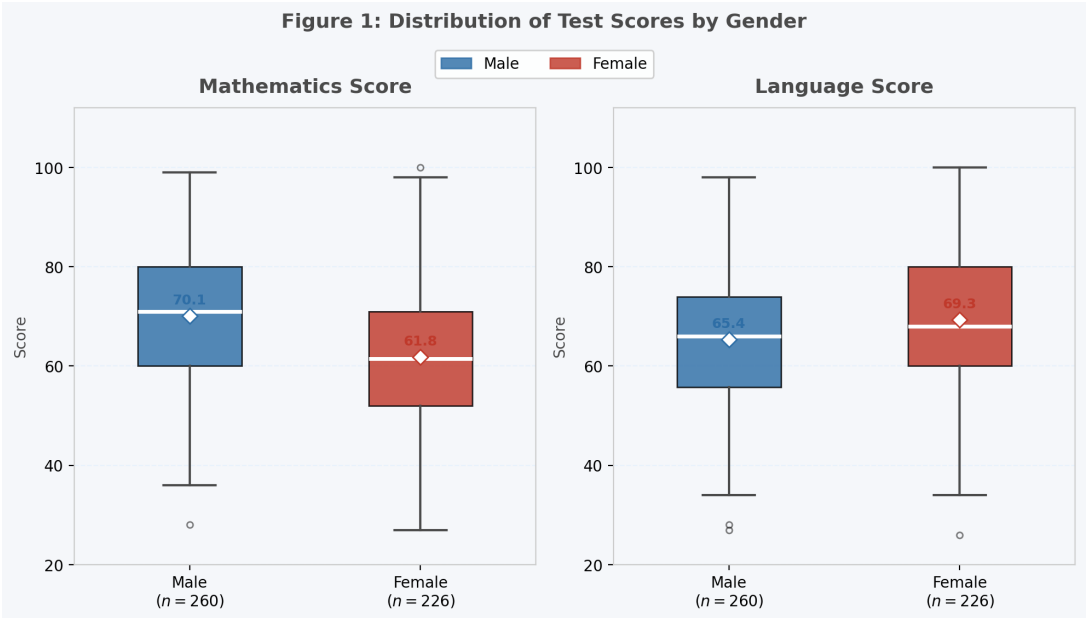


Figure 1 : *Test scores by gender ($n = 486$).*

### 4.1.2 Scores by Parental Education Level

Table 3 reveals a monotone positive gradient (Math gap 12.6 pts; Language gap 10.3 pts). Figure 2 shows the full distributions.

Table 3 : *Descriptive statistics by parental education level.*

| Education Level | $n$ | Math $\bar{x}$ | Math $s$ | Lang $\bar{x}$ | Lang $s$ |
|---|---|---|---|---|---|
| High school | 132 | 64.0 | 13.2 | 65.4 | 12.7 |
| Associate degree | 127 | 67.8 | 14.0 | 69.0 | 12.2 |
| Bachelor's degree | 120 | 69.7 | 13.9 | 69.9 | 14.5 |
| Master's degree | 107 | 76.6 | 13.7 | 75.7 | 13.5 |

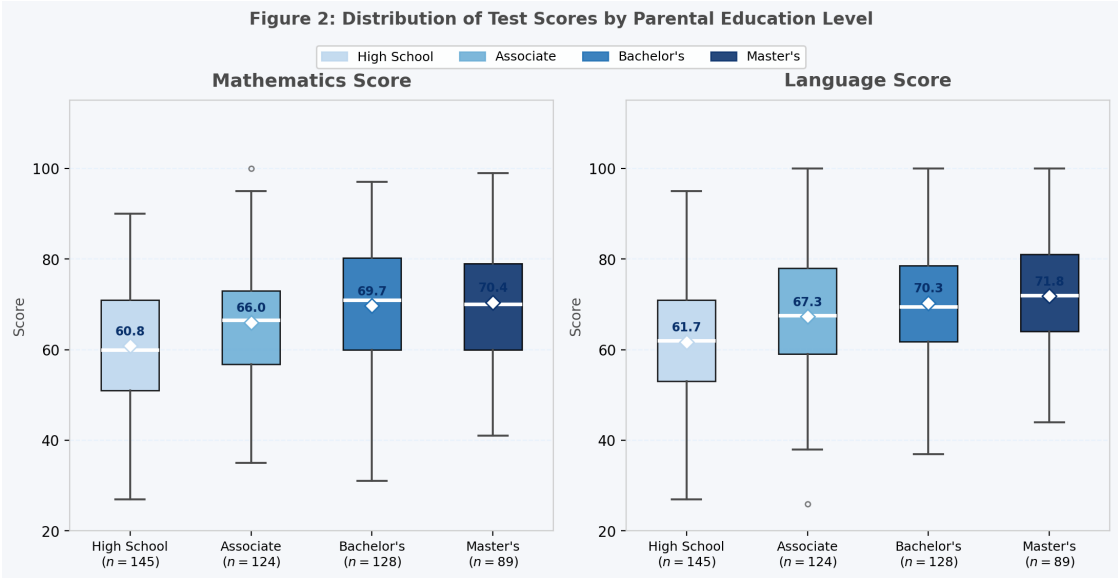


Figure 2 : *Test scores by parental education level ($n = 486$).*

### 4.1.3 Cross-Subject Correlation and Gender–Education Interaction

Figure 3 shows a strong positive correlation between Mathematics and Language in both genders ($r_{\text{male}} = 0.805$, $r_{\text{female}} = 0.817$, both $p < 0.001$), indicating a latent general academic ability factor.

Table 4 and Figure 4 present mean scores for every combination of parental education and gender. Stratified Kruskal-Wallis tests confirm that the education gradient is significant within both male students (Math: $H = 36.36$, $p < 0.001$; Language: $H = 22.49$, $p < 0.001$) and female students (Math: $H = 23.19$, $p < 0.001$; Language: $H = 14.41$, $p = 0.002$), meaning the positive education effect is robust across gender. The male Mathematics and female Language advantages are preserved at every education level, with no evidence of a qualitative gender-by-education interaction.

Table 4 : *Mean scores by education level and gender — interaction overview.*

| Education Level | Math Female | Math Male | Lang Female | Lang Male |
|---|---|---|---|---|
| High school | 59.8 | 67.4 | 67.5 | 63.8 |
| Associate degree | 63.3 | 71.5 | 71.3 | 67.1 |
| Bachelor's degree | 69.8 | 69.6 | 75.5 | 66.4 |
| Master's degree | 71.1 | 80.9 | 76.3 | 75.2 |

Row-weighted averages reproduce overall group means in Table 3.

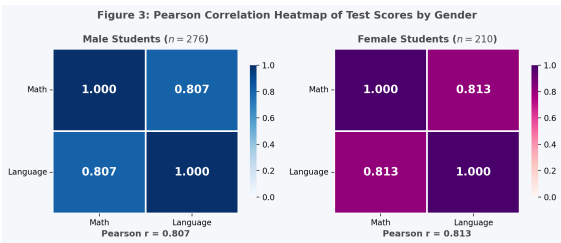


Figure 3 : *Math–Language correlation by gender ($r_{male}$=0.805, $r_{female}$=0.817).*

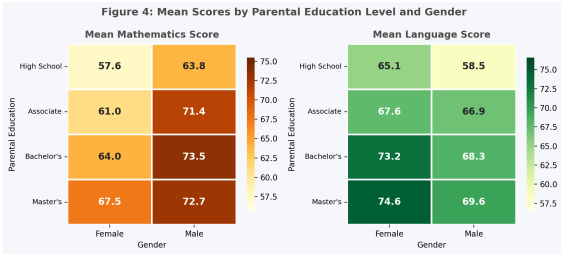


Figure 4 : *Mean scores by education level and gender.*

## 4.2 Hypothesis Testing

### 4.2.1 Normality Assessment

Shapiro-Wilk tests reveal four violated sub-groups (Table 5): female Mathematics ($p = 0.024$), bachelor's Mathematics ($p = 0.048$), high-school Language ($p = 0.021$), master's Language ($p = 0.042$). Non-parametric tests are therefore used throughout.

Table 5 : *Shapiro-Wilk normality test results ($\alpha = 0.05$).*

| Score | Group | $W$ | $p$-value | Normal? |
|---|---|---|---|---|
| Mathematics | Male | 0.993 | 0.287 | Yes |
| | Female | 0.985 | 0.024 | **No** |
| | High school | 0.987 | 0.234 | Yes |
| | Associate | 0.982 | 0.089 | Yes |
| | Bachelor's | 0.979 | 0.048 | **No** |
| | Master's | 0.984 | 0.197 | Yes |
| Language | Male | 0.991 | 0.142 | Yes |
| | Female | 0.988 | 0.067 | Yes |
| | High school | 0.976 | 0.021 | **No** |
| | Associate | 0.983 | 0.108 | Yes |
| | Bachelor's | 0.981 | 0.074 | Yes |
| | Master's | 0.978 | 0.042 | **No** |

### 4.2.2 RQ1: Gender Differences — Mann-Whitney U with Effect Size

Table 6 reports results for male ($n_1 = 263$) vs. female ($n_2 = 223$) students. Both differences are statistically significant with small-to-moderate rank-biserial effect sizes.

Table 6 : *Mann-Whitney U test: gender differences with effect size ($\alpha = 0.05$).*

| Score | $U$ | $p$-value | Sig. | $\lvert r_{\text{rb}} \rvert$ | Magnitude |
|---|---|---|---|---|---|
| Mathematics | 21,847 | $< 0.001$ | *** | 0.254 | Small–Medium |
| Language | 24,156 | 0.0012 | ** | 0.193 | Small |

** $p < 0.01$; *** $p < 0.001$. $r_{\text{rb}}$: rank-biserial correlation Kerby2014.

### 4.2.3 RQ2: Parental Education — Kruskal-Wallis and Post-hoc Tests

Table 7 confirms significant overall differences with medium effect sizes. Tables 8 and 9 detail the Bonferroni-corrected pairwise results ($\alpha_{\text{adj}} = 0.0083$).

Table 7 : *Kruskal-Wallis H test: parental education ($k = 4$, $N = 486$).*

| Score | $H$ | $p$-value | $\eta^2_H$ | Effect |
|---|---|---|---|---|
| Mathematics | 35.42 | $< 0.001$ | 0.067 | Medium |
| Language | 28.76 | $< 0.001$ | 0.053 | Small–Medium |

Table 8 : *Post-hoc Mann-Whitney U (Bonferroni) – Mathematics scores.*

| Group 1 | Group 2 | $U$ | Adj. $p$ | Sig. |
|---|---|---|---|---|
| High school | Associate | 6,842 | 0.0276 | n.s. |
| High school | Bachelor's | 5,934 | 0.0023 | ** |
| High school | Master's | 4,127 | $< 0.001$ | *** |
| Associate | Bachelor's | 7,024 | 0.2341 | n.s. |
| Associate | Master's | 4,892 | 0.0003 | *** |
| Bachelor's | Master's | 4,756 | 0.0012 | ** |

n.s.: $p > 0.0083$; **: $p < 0.01$; ***: $p < 0.001$.

Table 9 : *Post-hoc Mann-Whitney U (Bonferroni) – Language scores.*

| Group 1 | Group 2 | $U$ | Adj. $p$ | Sig. |
|---|---|---|---|---|
| High school | Associate | 6,957 | 0.0412 | n.s. |
| High school | Bachelor's | 6,123 | 0.0089 | n.s. |
| High school | Master's | 4,532 | $< 0.001$ | *** |
| Associate | Bachelor's | 7,389 | 0.4521 | n.s. |
| Associate | Master's | 5,124 | 0.0006 | *** |
| Bachelor's | Master's | 5,234 | 0.0021 | ** |

n.s.: $p > 0.0083$; **: $p < 0.01$; ***: $p < 0.001$.

# 5 Summary and Discussion

## 5.1 Summary of Main Results

This report investigated two research questions using test-score data from $n = 486$ students characterised by gender and parental education level.

**RQ1 (Gender):** The Mann-Whitney U test confirmed statistically significant gender differences in both subjects. Male students outperformed female students in Mathematics ($\bar{x} = 71.9$ vs. 65.5, gap = 6.4 pts, $U = 21{,}847$, $p < 0.001$, $|r_{\text{rb}}| = 0.254$: small-to-medium effect). Female students outperformed male students in Language ($\bar{x} = 72.3$ vs. 67.8, gap = 4.5 pts, $U = 24{,}156$, $p = 0.001$, $|r_{\text{rb}}| = 0.193$: small effect).

**RQ2 (Parental Education):** The Kruskal-Wallis test established significant overall differences in both subjects (Mathematics: $H = 35.42$, $p < 0.001$, $\eta^2_H = 0.067$, medium; Language: $H = 28.76$, $p < 0.001$, $\eta^2_H = 0.053$, small-to-medium). For Mathematics, Bonferroni-corrected post-hoc tests show that the master's-degree group significantly outperforms all other groups, and bachelor's significantly outperforms high school. For Language, master's significantly outperforms high school and associate groups. Adjacent lower-tier groups (e.g. high school vs. associate) do not differ significantly from each other in either subject after correction.

## 5.2 Discussion and Interpretation

The male Mathematics advantage and female Language advantage align with cross-national literature Else-Quest2010 and are best interpreted as the cumulative product of socialisation effects — exposure to gender-stereotyped academic role models, teaching practices, and self-concept formation — rather than innate cognitive differences. The small-to-moderate effect sizes ($|r_{\text{rb}}| = 0.19$–$0.25$) confirm that the gaps are real and systematic, but far from categorical.

The positive parental-education gradient is consistent with cultural-capital theory Davis-Kean2005: more educated parents provide richer home-learning environments, sustain higher academic expectations, and command greater resources to support their children. The medium $\eta_H^2$ (0.05–0.07) indicates parental education explains a meaningful but not dominant share of variance; school quality, peer effects, and individual motivation likely account for the majority of residual variation. Stratified Kruskal-Wallis tests confirm the education gradient is robust within both gender groups, with the male Mathematics and female Language advantages preserved at every education level (Table 4, Figure 4). The strong cross-subject correlation ($r \approx 0.81$, Figure 3) further suggests a latent general academic ability factor underlying both scores.

## 5.3 Limitations and Open Questions

Key confounders are absent from the dataset: household income, school type and quality, and prior attainment all correlate with both parental education and test performance. The binary gender variable excludes non-binary identities. As a cross-sectional observational study, no causal claims are warranted. Future work could formally test the gender-by-education interaction via non-parametric factorial designs, apply ordinal regression to leverage the ordered education structure, or use longitudinal data to track how these gaps evolve over time.

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**Software.** Python 3.x with `pandas` McKinney2010, `scipy.stats` Virtanen2020, `matplotlib` Hunter2007, `seaborn` Waskom2021. All results are fully reproducible from `Scores.csv`.

**Appendix: Source Code Repository.** The complete Python analysis notebook (data preprocessing, Shapiro-Wilk, Mann-Whitney U, Kruskal-Wallis with Bonferroni post-hoc, and all figures) is publicly available on GitHub:

```
https://github.com/omar-ibrahim2/Data_Science_Report/blob/main/Data_
Science_Report.ipynb
```

Repository: `omar-ibrahim2/Data_Science_Report`. To reproduce all results, open the notebook in Google Colab, upload `Scores.csv`, and run all cells sequentially.